ARJUNA JEE 2027

Lecture -11 | Mathematics

PERMUTATIONS AND COMBINATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Circular Permutation**

B **Derangement**

C **Question Practice**

D



H.W. Discussion

Acidic Question

How many words of seven letters can be formed by using the letters of the word 'SUCCESS' so that neither the two C nor any two S are together.

A 96

B 41

C 264

D 40

S, S, S, C, C, U, E

_ C _ C _ U _ E _

$\frac{4!}{2!} \times {}^{5}C_3 \times 1$

S C S C U S E ✓

C S C U S E S ✓

S C C S U S E ✗

S, S, S

_ CC _ U _ E _ → ${}^{4}C_3 \times 1 \times 3!$

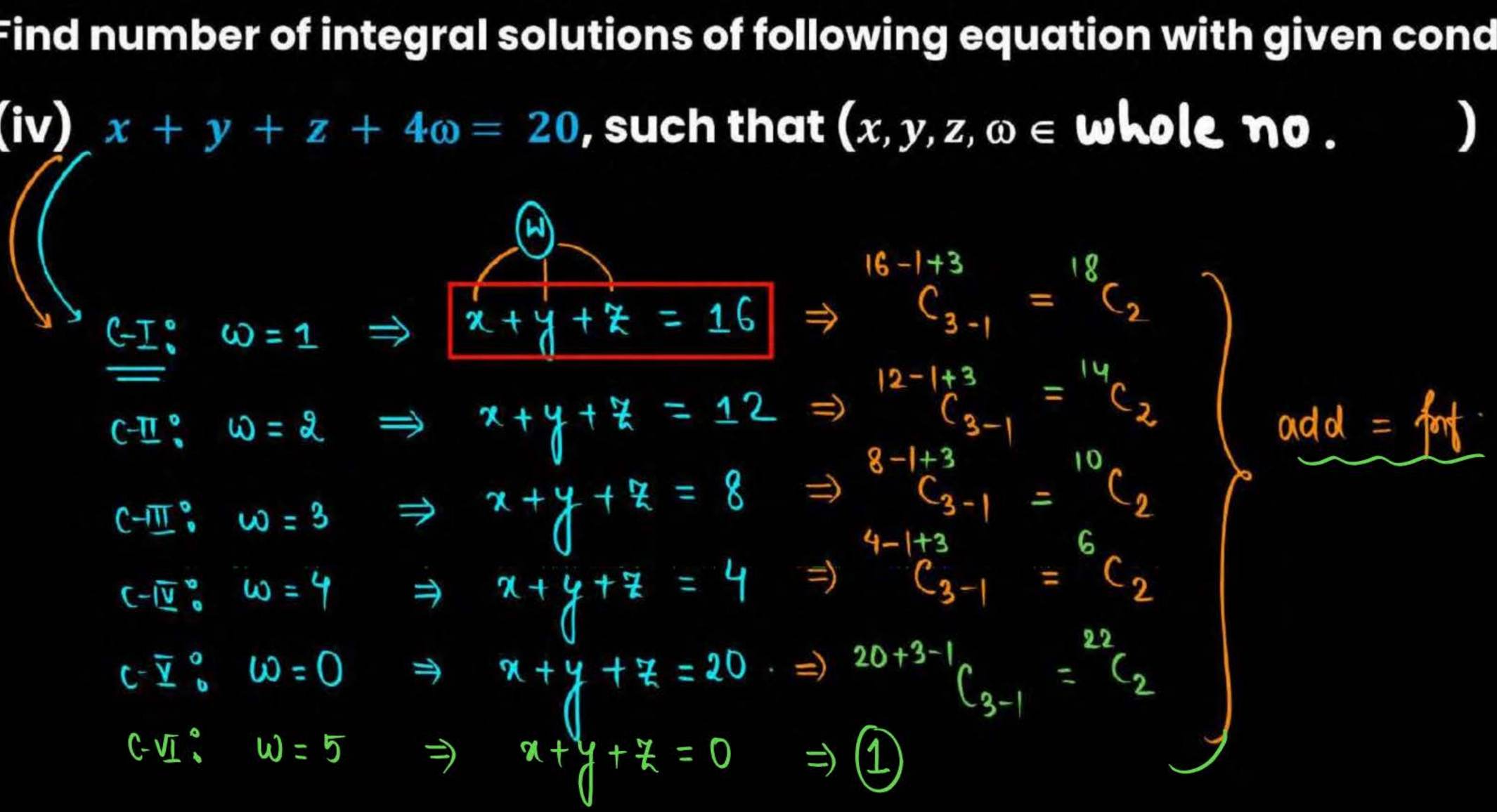
-C _ C _ U _ E _ Required.

$fnf = \frac{4!}{2!} \times 10 - 4 \times 6$

$= 120 - 24 = 96$

Find number of integral solutions of following equation with given conditions:

(iv) $x + y + z + 4\omega = 20$, such that ($x, y, z, \omega \in$ whole no.)



Find number of integral solutions of following equation with given conditions:

(v) $x + y + z \leq 20$, **such that** ($x, y, z \in$ **non-negative integers**)

(Whole No.)

ex:- $10 + 0 + 10$ ✓

$5 + 6 + 0$ ✓

$1 + 1 + 1$ ✓

$0 + 0 + 0$ ✓

False beggar

$x + y + z + t = 20$

10 0 7	3
5 1 4	10
0 0 0	20
9 9 2	0

all possible $= {}^{n+r-1}C_{r-1}$

$= {}^{20+4-1}C_{4-1}$

$= {}^{23}C_3$ ✓

Find number of integral solutions of following equation with given conditions:

(vi) $35 < x + y + z \leq 52$, **such that** ($x, y, z \in$ **positive integers**)

OR

$36 \leq x + y + z \leq 52$

$x + y + z \leq 52$

$\underbrace{x + y + z}_{N} + \underbrace{t}_{W} = 52$

$x' + y' + z' + t = 49.$

all possible $= {}^{49+4-1}C_{4-1} = {}^{52}C_3.$

also counted cases when $x + y + z \leq 35$

$x' + y' + z' + t = 32$

$= {}^{32+4-1}C_{4-1} = {}^{35}C_3.$

$f_{nf} = {}^{52}C_3 - {}^{35}C_3$

Maha Bumper Question

Find number of integral solutions of following equation with given conditions:

(vii) $xyz = 2^4 \cdot 3^3 \cdot 5^2$, $(x, y, z \in N)$

2222 333 55

2 id → 3 child → $p+q+r=2$ → $^{2+3-1}C_{3-1} = {}^{4}C_2$

1 0 1

2 0 0

3 id. → 3 child → $p+q+r=3$ → $^{3+3-1}C_{3-1}$ (${}^{5}C_2$.

1 1 1

0 2 1.

1 0 2

4 id. things dist. to x, y, z (3 child.)

$p+q+r=4$ → all possible soln. $= {}^{4+3-1}C_{3-1} = {}^{6}C_2$.

0 1 3

2 1 1

0 0 4

⋮

$\text{Inf} = ({}^{6}C_2 \times {}^{5}C_2 \times {}^{4}C_2)$

ex:-

x	y	z
35	23	22235
2255	233	23
3×1	1	22225533

DIBY - (X)

$\# \; xyz = \underline{2}^{4} \cdot 3^{5} \cdot \underline{10}^{4} \quad (x, y, z \in N)$

$\underline{10} = \underline{2 \cdot 5}$

$\underline{2}^{4} \cdot 5^{4}$

OR

$\# \; xyz = 2^{8} \cdot 3^{5} \cdot 5^{4}$

Question

Notes!!

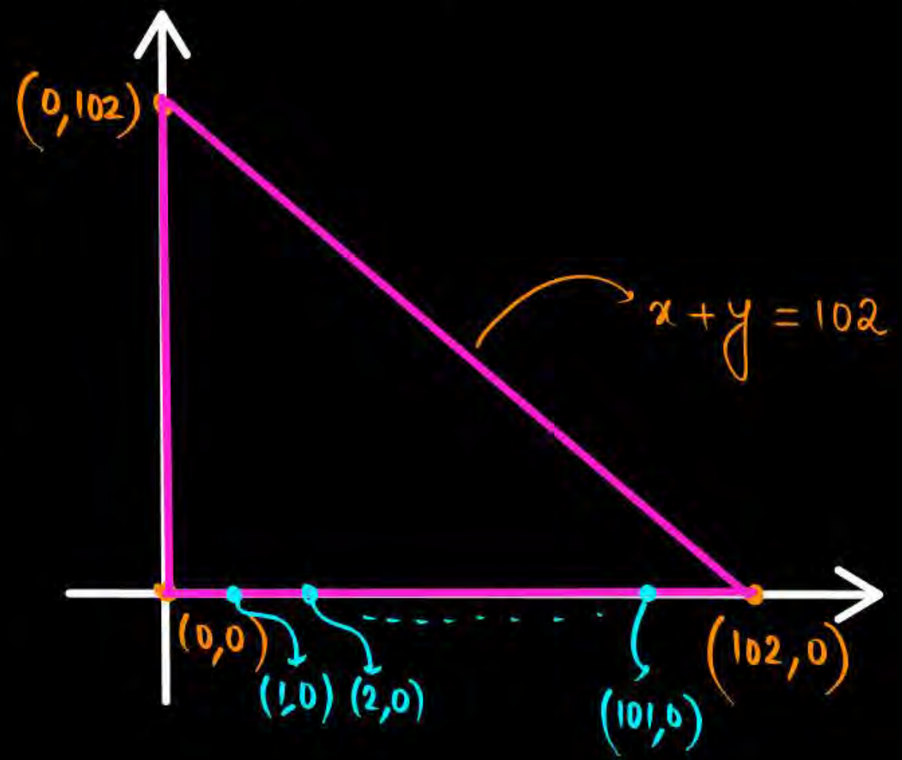
Number of points with integral co-ordinates that lie strictly inside the triangle whose vertices are (0, 0), (0, 102) and (102, 0) is.

A 5050

B 5151

C 4950

D 5253



H-1

$x + y < 102$, $x > 0, y > 0$

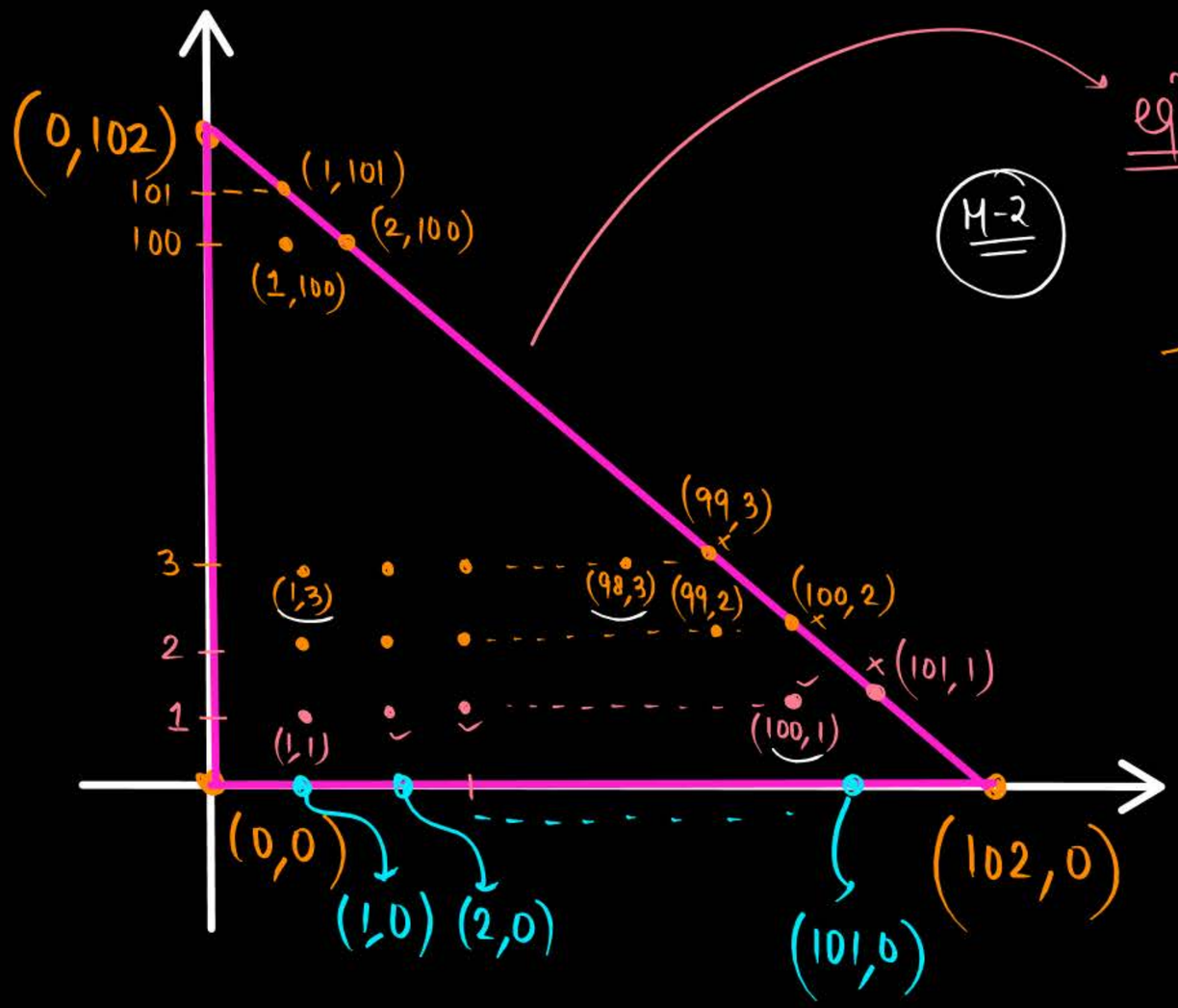
$x + y \leq 101$, $x, y \in N$

$x + y + t = 101$ $\quad 5050 = \frac{101 \times 100}{2}$

$x' + y' + t = 99$

$99 + 3 - 1 \quad 101$

$C_{3-1} = {}^{101}C_2$



M-2

eqⁿ : $x + y = 102$

Total $= 100 + 99 + 98 + \ldots\ldots + 1$

$= \frac{100 \times 101}{2}$

$= 50 \times 101$

$= 5050.$

Point (x, y) where x & y both integer then point is called lattice point.

ex:- $(3, 4)$

$(1, 2)$

$(-1, 0)$

$(5, -7)$

⋮

Question [JEE Mains-2023]

[Ans. : (D)]

The number of triplets (x, y, z) where x, y, z are distinct non negative integers satisfying $x + y + z = 15$, is



A 136

B 80

C 92

D 114

NON-CONSECUTIVE SELECTION

Question

Consider set S = {1, 2, 3, 4, 5, 6, 7, 8, 9}, number of ways of selecting 4 numbers such that no two are consecutive.

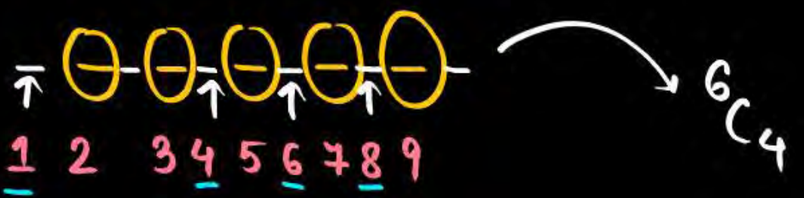
ex: 1,5,7,9 ✓

2,4,8,1 ✗

2,5,7,9 ✓

1,4,6,9 ✓

⋮



NON-CONSECUTIVE SELECTION

Question

Consider set S = {1, 2, 3, 4, 5, 6, 7, 8, 9}, number of ways of selecting 4 numbers such that no two are consecutive.

$n = 9$, $r = 4$ $\rightarrow$ $^{9-4+1}C_4$

ex:- 1, 3, 6, 9

2, 5, 7, 9

1, 3, 5, 8

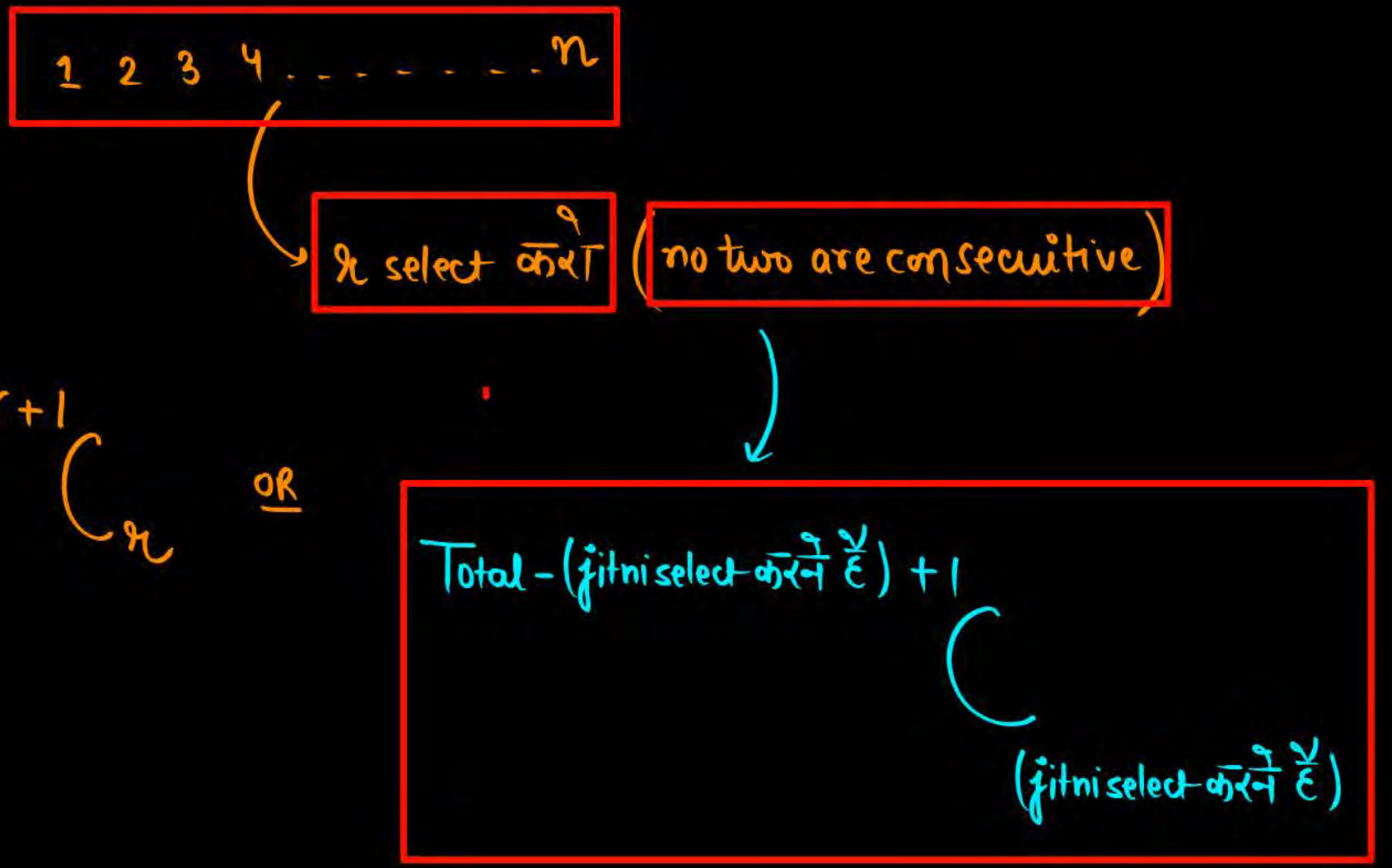
2, 4, 6, 9.

$x_1 + x_2 + x_3 + x_4 + x_5 = 5$

$\geq 0 \quad \geq 1 \quad \geq 1 \quad \geq 1 \quad \geq 0$

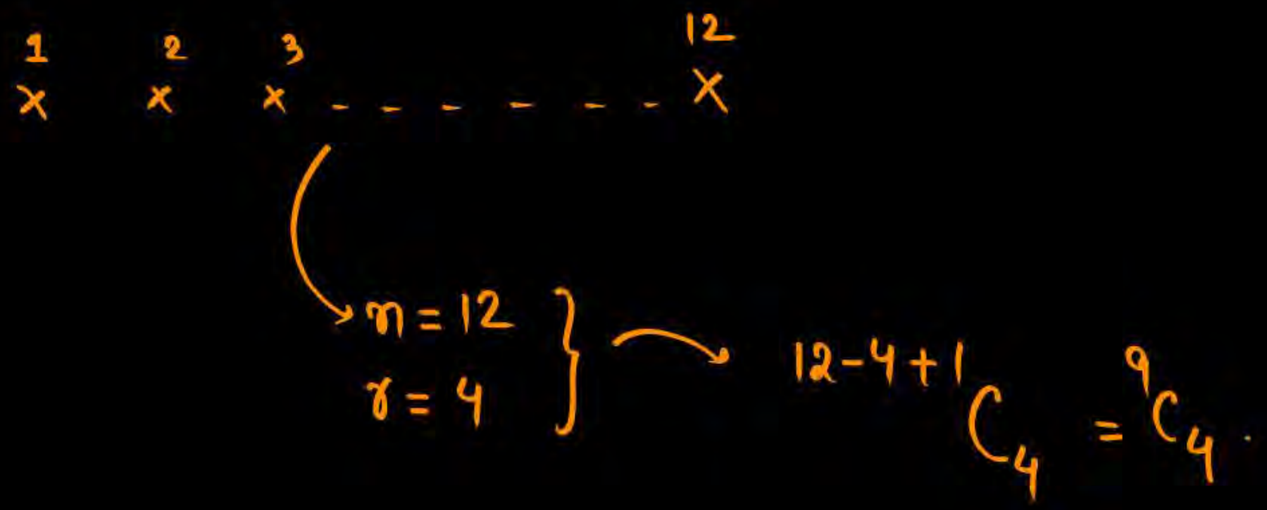
$x_1 + x_2' + x_3' + x_4' + x_5 = 2$

$2+5-1C_{5-1} = {}^6C_4$


1 2 3 4 . . . . . n
r select करने हैं
(no two are consecutive)
$^{n-r+1}C_r$
OR
Total - (jitni select करने हैं) + 1
C
(jitni select करने हैं)

Question

A train having 12 stations enroute has to be stopped at 4 stations. Number of ways it can be stopped if no two of the stopping stations are consecutive.



Question [JEE (Adv.)-2020, Paper-2]

An engineer is required to visit a factory for exactly four days during the first 15 days of every month and it is mandatory that no two visits take place on consecutive days. Then the number of all possible ways in which such visits to the factory can be made by the engineer during 1-15 June 2021 is _____.

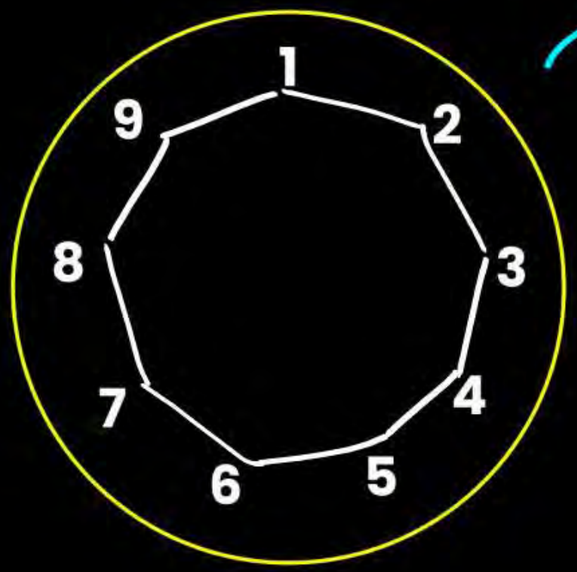
June → 1 2 3 4 15

$^{15-4+1}C_4 = {}^{12}C_4$ fnf.

NON-CONSECUTIVE SELECTION

Question

If these nine numbers are placed on a circle (as shown), number of ways of selecting 4 numbers such that no two are consecutive.



ex :- 1,5,7,9 ✗

9,3,5,7 ✓

4,6,8,2 ✓

⋮

HI

dine में assume krke ans.

⇓

$^{9-4+1}C_4$ − (when end point 1 & 9 selected)

$^{5-2+1}C_2$

$^{6}C_4 - {}^{4}C_2 = 15 - 6 = 9$

NON-CONSECUTIVE SELECTION

Question

If these nine numbers are placed on a circle (as shown), number of ways of selecting 4 numbers such that no two are consecutive.



ex :- 1, 5, 7, 9 ✗

9, 3, 5, 7 ✓

4, 6, 8, 2 ✓

⋮

M-2

$$\underbrace{{}^{9}C_{1} \times {}^{6-3+1}C_{3}}_{4} = \frac{9 \times {}^{4}C_{3}}{4} = 9$$



NON-CONSECUTIVE SELECTION

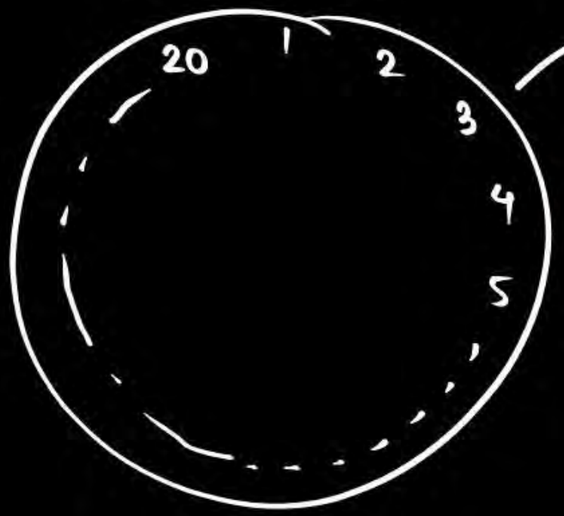
Number of ways of selecting 'r' – things out of 'n' – consecutive things Placed in a line such that no two things are consecutive = $^{n-r+1}C_r$

Number of ways of selecting 'r' – things out of 'n' – consecutive things Placed on a circle such that no two things are consecutive

$$= \frac{^{n}C_1 \cdot {}^{n-r-1}C_{r-1}}{r} \quad \text{or} \quad \boxed{^{n-r+1}C_r} - \boxed{^{n-r-1}C_{r-2}}$$

Number of octagons (8 sided polygon) that can be formed by joining vertices of Icosagon (20 sided polygon) such that no side of Octagon is common with polygon.

$n = 20$



select any r (8) (no two are consecutive)

M-1

$${}^{n-r+1}C_{r} - {}^{n-r-1}C_{r-2}$$

$${}^{20-8+1}C_{8} - {}^{20-8-1}C_{8-2} = {}^{13}C_{8} - {}^{11}C_{6}$$

M-2

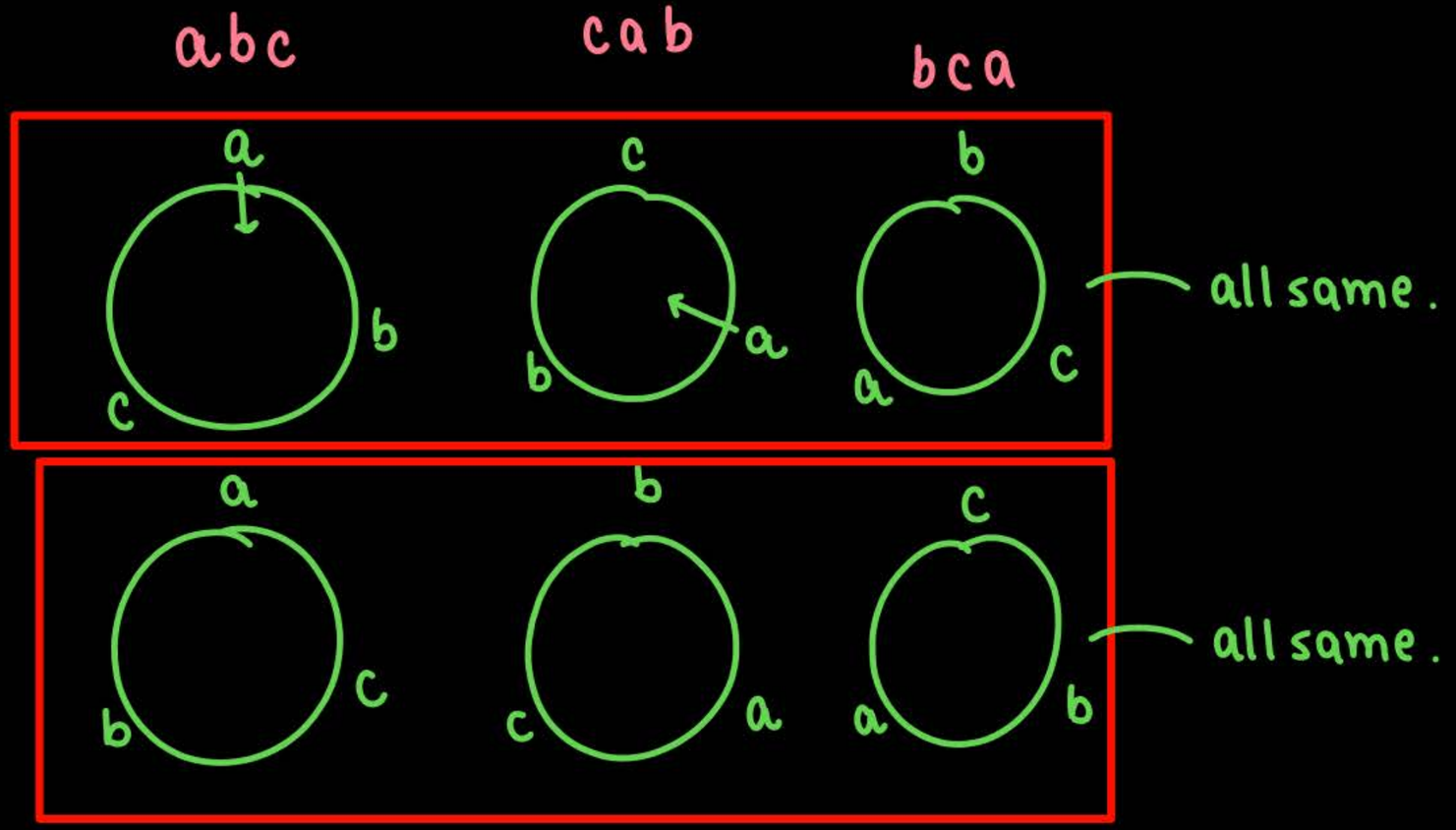
$${}^{20}C_{1} \times {}^{20-8-1}C_{8-1}$$

$$8$$

• CIRCULAR

abc ✓
acb
bca ✓
bac
cab ✓
cba

(6) = 3!



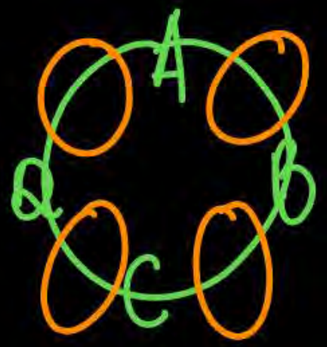
CIRCULAR PERMUTATION

If we arrange things on Closed curve then it is called as **CIRCULAR PERMUTATTION**.

NOTE:

1. Closed curve has no starting & ending point. ✓
2. Number of gaps along 'n' things on circle are 'n'. ✓

_A_B_C_D_
↳ Gap = 5



Gap = 4.

TWO FORMULA's

(i) Number of ways in which n-different things can be arranged on Circle = $(n-1)!$

*** (ii) Number of ways in which n-different things can be arranged on Circle when Clockwise & Anticlockwise arrangements are considered as same $= \frac{(n-1)!}{2}$



In case of NECKLACE & GARLAND

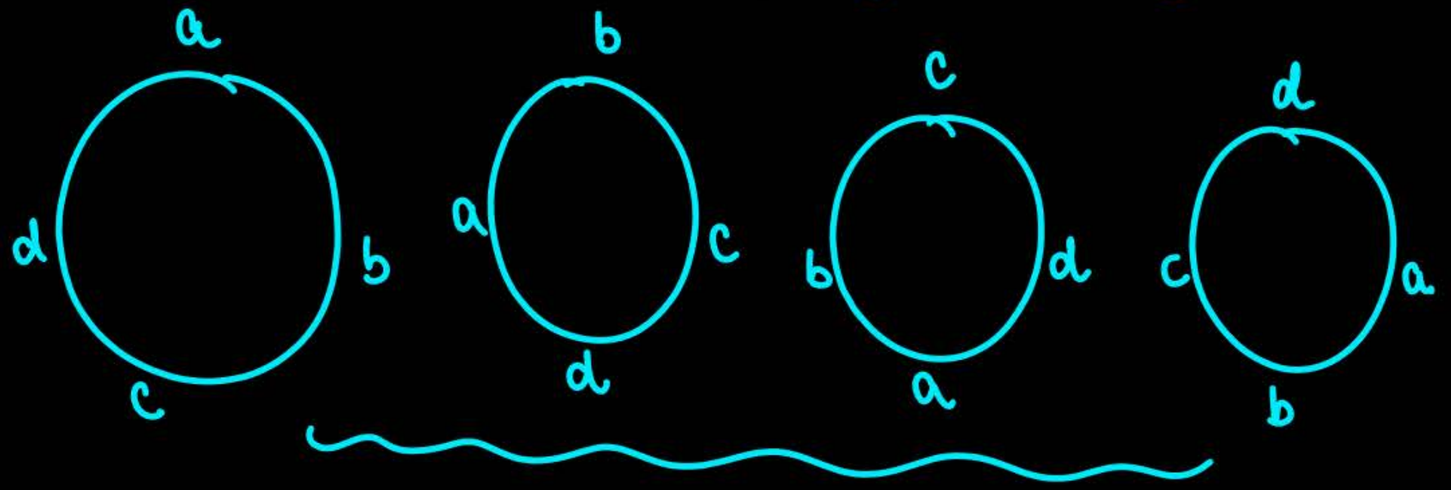
In case when all have different NEIGHBOURS in every arrangement.

$24 = 4!$

abcd
bcda
cdab
dabc
⋮

Circle

$(4-1)! = 3! = 6$



same.

Question

In how many ways 5 boys can be seated around a circular table such that they don't have same neighbours. (in any arr.)

$(5-1)! = 4!$

fnf $= \frac{(5-1)!}{2} = \frac{4!}{2} = 12$

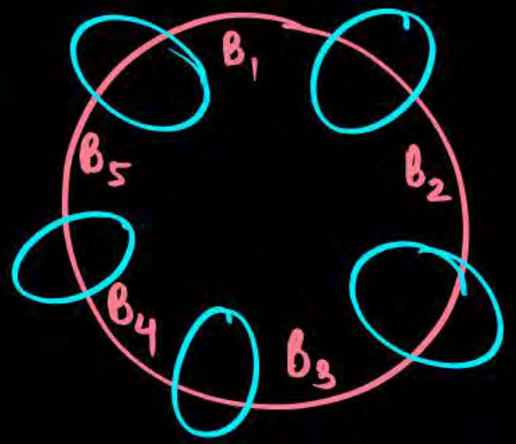
Question

How many garlands can be made using 6 different flowers out of 10 different available flowers.

$$^{10}C_6 \times \frac{(6-1)!}{2}$$

Example

Find number of ways in which 5B and 5G can be seated on a circle alternately.



(A) $(5-1)!_0 \times (5-1)!_0$ ✗

(B) $\boxed{(5-1)!_0 \times 5!_0}$ ✓

Example

Find number of ways in which 5B and 5G can be seated on a circle

Such that all boys are together.

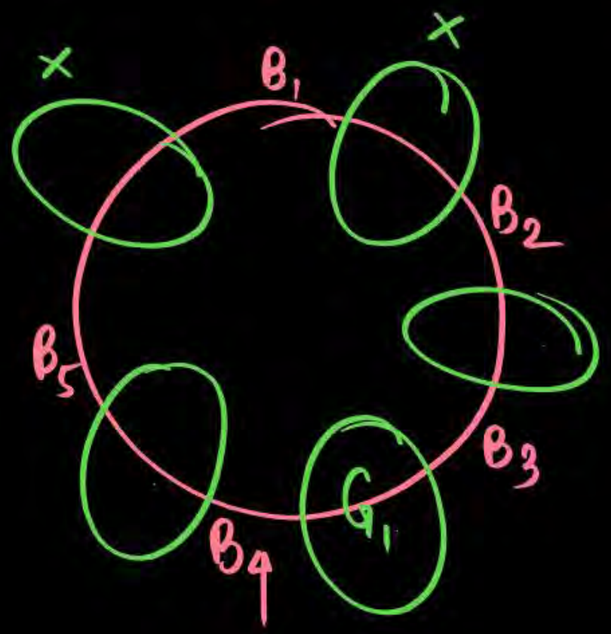
$B_1\ B_2\ B_3\ B_4\ B_5$, G_1, G_2, G_3, G_4, G_5



$$(6-1)! \times 5!$$

Example

Find number of ways in which 5B and 5G can be seated on a circle alternately if a particular B_1 and G_1 are never adjacent to each other in any arrangement.



$$(5-1)!_0 \times {}^3C_1 \times 1 \times 4!_0$$

fnf.

Question [JEE Mains-2023]

The number of ways, in which 5 girls and 7 boys can be seated at a round table so that no two girls sit together, is



A 720

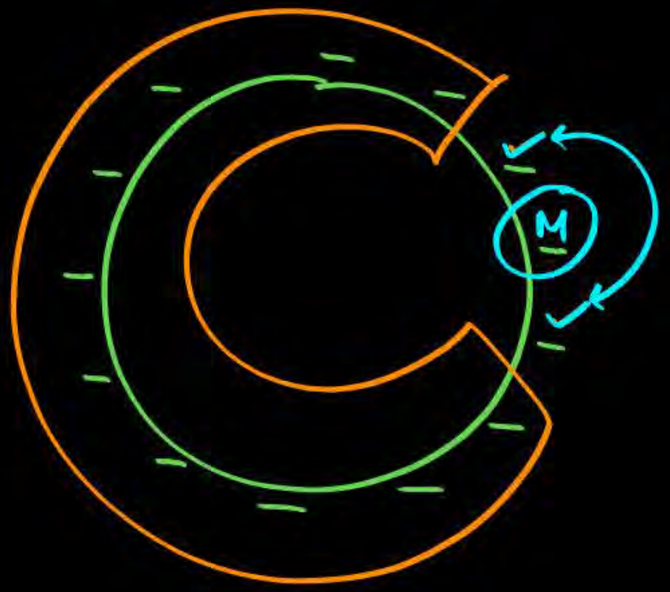
B $126(5!)^2$

C $7(360)^2$

D $7(720)^2$

Example

The 10 students of batch B feel they have some conceptual doubt on "Circular permutation". Mr. Mathew called them in discussion room and asked them to sit down around a circular table which is surrounded by 13 chairs. Mr Mathew told that his adjacent seat should not remain empty. The number of ways, in which the students can sit around a round table if Mr. Mathew also sit around a chair.



$$1 \times {}^{10}C_2 \times 2! \times {}^{10}C_8 \times 8!$$

DO IT BY YOURSELF (DIBY)

DIBY-58

In an examination of Mathematics paper, there are 20 questions of equal marks and the question paper is divided into three sections: A, B and C. A student is required to attempt total 15 questions taking at least 4 questions from each section. If section A has 8 questions, section B has 6 questions and section C has 6 questions, then the total number of ways a student can select 15 questions is [JEE Mains-2024]

[Ans. : 11376]

DIBY-59

In how many ways 5 NEET & 5 JEE Aspirants can be seated

(a) All JEE aspirants are together

(b) All NEET aspirants are together & all JEE aspirants are together

(c) Not all JEE aspirants are together

(d) No 2 JEE aspirants are together.

[Ans. : .....]

DIBY-60

Find the number of all 6 digit numbers such that all the digits of each number are selected from the set $\{1, 2, 3, 4, 5\}$ and any digit that appears in the number appears at least twice.

[Ans. : 1405]

DIBY-61

The number of words that can be formed using all the letter of the word "CURRICULUM" in which two U's are together but separated from the third 'U' in $K(7!)$. The value of K is

(A) 7 (B) 9 (C) 12 (D) 14

[Ans. : (D)]

DIBY-62

The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is [IIT-JEE-2012]

(A) 75 (B) 150 (C) 210 (D) 243

[Ans. : (B)]

DIBY-63

The students $S_1, S_2, \ldots, S_{10}$ are to be divided into 3 groups A, B and C such that each group has at least one students and the group C has at most 3 students. Then the total number of possibilities of forming such groups is

[JEE Mains-2021]

[Ans. : 31650]

DIBY-64

The total number of three-digit numbers, divisible by 3, which can be formed using the digits 1, 3, 5, 8 if repetition of digits is allowed, is [JEE Mains-2023]

(A) 21 (B) 22 (C) 18 (D) 20

[Ans. : (B)]

DIBY-65

Let the set $S = \{2, 4, 8, 16, \ldots. 512\}$ be partitioned into 3 sets A, B, C with equal number of elements such that $A \cup B \cup C = S$ and $A \cap B = B \cap C = A \cap C = \phi$. The maximum number of such possible partitions of S is equal to [JEE Mains-2024]

(A) 1640 (B) 1520 (C) 1710 (D) 1680

[Ans. : (D)]

DIBY-66

A natura number has prime factorization given by $n = 2^x 3^y 5^z$, where y and z are such that $y + z = 5$ and $y^{-1} + z^{-1} = \frac{5}{6}, y > z$. Then the number of odd divisors of n, including 1, is [JEE Mains-2021]

(A) 11 (B) $6x$ (C) 12 (D) 6

[Ans. : (C)]

DIBY-67

Number of integral solutions to the equation $x + y + z = 21$, where $x \geq 1$, $y \geq 3$, $z \geq 4$, is equal to [JEE Mains-2023]

[Ans. : 105]

DIBY-68

There are infinite number of Identical BLUE, RED, GREEN and YELLOW balls available for selections. In how many ways can we select 10 balls.

(A) ${}^{14}C_3$ (B) ${}^{13}C_{10}$ (C) ${}^{12}C_{10}$ (D) ${}^{14}C_4$

[Ans. : (B)]

DIBY-69

The number of triangles whose vertices are at the vertices of a regular octagon but none of whose sides is a side of the octagon is [JEE Mains-2024]

(A) 56 (B) 16 (C) 24 (D) 48

[Ans. : (B)]

DIBY-70

The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in row so that between any two red cubes, there should be at least 2 blue cubes, is [JEE Mains-2022]

[Ans. : 56]

DIBY-71

Number of ways of arranging 8 identical books into 4 identical shelves where any number of shelves may remain empty is equal to [JEE Mains-2024]

(A) 18 (B) 16 (C) 12 (D) 15

[Ans. : (D)]

DIBY-72

The total number of positive integral solutions (x, y, z) such that $xyz = 24$ is

[JEE Mains-2021]

(A) 36 (B) 45 (C) 24 (D) 30

[Ans. : (D)]

DIBY-73

Let $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$, then

[JEE Mains-2024]

(A) $\alpha \in$ N and $\beta \in$ N

(B) $\alpha \in$ N and $\beta \notin$ N

(C) $\alpha \notin$ N and $\beta \in$ N

(D) $\alpha \notin$ N and $\beta \notin$ N

[Ans. : (A)]

DIBY-74

From a group of 7 batsmen and 6 bowlers, 10 players are to be chosen for a team, which should include atleast 4 batsmen and atleast 4 bowlers. One batsmen and one bowler who are captain and vice-captain respectively of the team should be included. Then the total number of ways such a selection can be made, is [JEE Mains-2025]

(A) 165 (B) 155 (C) 145 (D) 135

[Ans. : (B)]

DIBY-75

The number of ways, in which the letters A, B, C, D, E can be placed in the 8 boxes of the figure below so that no row remains empty and at most one letter can be placed in a box, is [JEE Mains-2025]

(A) 5880 (B) 960 (C) 840 (D) 5760



[Ans. : (D)]

DIBY-76

The number of functions f from $\{1, 2, 3, ..., 20\}$ only $\{1, 2, 3, ..., 20\}$ such that $f(k)$ is a multiple of 3, whenever k is a multiple of 4, is [JEE Mains-2019]

(A) $6^5 \times (15)!$ (B) $5! \times 6!$ (C) $(15)! \times 6!$ (D) $5^6 \times 15$

[Ans. : (C)]

DIBY-77

The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can placed in a row so that between any two red cubes there should be at least 2 blue cubes, is [JEE Mains-2022]

[Ans. : 56]

DIBY-78

For $n \geq 2$, let S_n denote the set of all subsets of $\{1, 2, ..., n\}$ with no two consecutive numbers. For example $\{1, 3, 5\} \in S_6$, but $\{1, 2, 4\} \notin S_6$. Then $n(S_5)$ is equal to [JEE Mains-2025]

[Ans. : 13]

DIBY-79

The largest $n \in N$ such that 3^n divides 50! is [JEE Mains-2025]

(A) 21 (B) 22 (C) 20 (D) 23

[Ans. : (B)]

DIBY-80

In an examination 5 students have been allotted their seats as per their roll numbers. The number of ways, in which none of the students sits on the allotted seat, is [JEE Mains-2023]

[Ans. : 44]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians